

Field study on the effect of vegetation on the performance of soil methanotrophy-based engineered systems – Column experiments

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ABSTRACT

Current literature provides conflicting information on the role vegetation plays when considering methane (CH₄) oxidation potential of engineered Biosystems, such as landfill biocovers (LBCs), bio-windows and methane biofilters. The primary objective of this study was to determine whether the impact of vegetation on biological CH₄ oxidation was positive or negative and to explain the reasons for the observations using a variety of experiments. A total of eight flow-through columns (two alfalfa, two native grass, two canola and two bare-soil replicates) were set up outdoors to simulate field operation in cold climatic conditions. Each column was layered with 18 cm of topsoil and 32 cm of compost mixture (compost 30%: topsoil 70% v/v) as packing material and treated with CH₄ fluxes ranging from 180 to 815 g CH₄ m⁻² d⁻¹. The bare-soil columns exhibited the highest CH₄ oxidation rate of 455 g CH₄ m⁻² d⁻¹, while the maximum CH₄ oxidation rates for the vegetated columns ranged between 147 and 171 g CH₄ m⁻² d⁻¹ with the alfalfa column showing the lowest. Gas profiles of vegetated columns showed high concentrations of nitrogen (N₂) and oxygen (O₂) at all depths, possibly due to increased permeability created by the plant root systems. Quantitative Polymerase Chain Reaction (q-PCR) assessment showed that *pmoA* gene copy numbers, indicative of methanotrophic population levels, were higher in bare-soil columns than in vegetated columns. The Illumina based sequencing of 16S rRNA gene showed that Type I methanotrophs dominated both vegetated and bare-soil columns. Soil incubation experiments conducted to determine oxidation kinetic parameters also indicated greater methanotrophic activity in the bare-soil columns than in vegetated columns. The plant data collected at the end of the column experiments provided clear evidence of CH₄ escape due to preferential pathways created by plant roots (reaching 38 cm, 31.5 cm and 26.5 cm by alfalfa, canola and native grass, respectively) that resulted in decreased CH₄ oxidation in vegetated columns. The study results clearly demonstrate that vegetation decreases CH₄ oxidation at high loading rates, notwithstanding the type of vegetation.

1. Introduction

Methane (CH₄) is a potent greenhouse gas (GHG) that has a global warming potential of 28–34 on a 100-year time horizon (Timma et al., 2020). Global CH₄ emissions are estimated at 550 Tg CH₄ yr⁻¹ (Kirschke et al., 2013) and emissions from the waste management sector accounts for about 8% of total global CH₄ emissions (Blanco et al., 2014). Fugitive

emission from sanitary landfills accepting biodegradable organic waste is the third highest anthropogenic source of CH₄ (Bian et al., 2018; Xin et al., 2016). Engineered systems that incorporate soil methanotrophy are the most cost-effective methods available to mitigate fugitive landfill CH₄ emissions (La et al., 2018). Such CH₄ oxidation based engineered systems include landfill biocover (LBC) as the final cover, bio-windows and methane biofilters (La et al., 2018; Scheutz et al., 2009). In these

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engineered systems, CH₄ is converted into carbon dioxide (CO₂) by CH₄ oxidizing bacteria, known as methanotrophs, which are planktonic or attached-growth (attached to the surface of soil or stone) naturally occurring aerobic bacteria usually present in the aerobic interfaces of locations containing high CH₄ gas concentrations, such as paddy fields and landfill cover soils (Reddy et al., 2019).

Although researchers have long understood the phenomenon of soil methanotrophy, the potential of its application to control fugitive CH₄ emissions has only recently been realised. Over the last three decades, considerable research has been undertaken to understand the fundamentals of soil methanotrophy and the factors affecting the CH₄ oxidation capacity of various engineered biosystems (La et al., 2018). Based primarily on laboratory batch or flow-through column experiments, researchers have determined the effects of key factors such as oxygen (O₂) and nutrient availability, moisture content, temperature, and pH on methanotrophic activity in granular media (Bohn et al., 2011; Reay et al., 2005). Although, the presence of vegetation is a critical factor that influences CH₄ oxidation capacity, considering the difficulty of conducting valid experiments in the laboratory and in the field, only a handful of studies have focused on determining the effects of vegetation on biological CH₄ oxidation (Ndanga et al., 2015).

Several lab-based flow-through column studies conducted in the past have indicated that the presence of plants in landfill cover soils could be beneficial as roots increase O₂ concentrations at deeper layers, providing ideal conditions for methanotroph growth and potentially increasing CH₄ oxidation (Bohn et al., 2011). The plant rhizosphere is considered a hot-spot for bacterial growth due to the wide range of substrates provided by root exudates (Stralis-Pavese et al., 2006). Other researchers postulated that the plant root system could potentially enhance CH₄ oxidation through reduction of soil moisture that could increase soil aeration and gas permeability (Reichenauer et al., 2011; Zhang et al., 2013). Reichenauer et al. (2011) examined gas profiles along the depths of columns and conducted batch experiments on samples obtained at different depths and concluded that plants have a positive influence on CH₄ oxidation. Hilger et al. (2000) conducted lab column experiments with artificial lighting to promote vegetation growth and observed 47% CH₄ oxidation efficiency in the grass-covered columns compared to only 37% in a bare-soil column. Stralis-Pavese et al. (2004) found that different types of plants affected the type and number of methanotrophs in soil, however the study did not compare vegetated and non-vegetated systems. In a study involving numerical modeling, lowest modeled surface CH₄ emissions were always observed in vegetated soil covers under all simulated climates (Abichou et al., 2015). Wang et al. (2008) conducted batch microcosm experiments to investigate the effects of *Chenopodium album* L. (i.e., a landfill gas tolerant plant) and observed a positive correlation between CH₄ oxidation and increase in microbial diversity within the rhizosphere of plants. However, it should be noted that the batch experiments might not represent the long-term performance under field conditions. Furthermore, most of these studies were conducted in laboratories under controlled conditions that did not represent field behaviour.

Although the above studies showed that plants positively impact CH₄ oxidation, other researchers have observed negative effects, primarily in dry conditions, as plant transpiration results in desiccation and lower microbial activity. A study conducted by Tanthachoon et al. (2008), used compost vegetated with *Panicum repens* and quantified a CH₄ oxidation rate of 12 mol CH₄ m⁻³ d⁻¹ during wet season, however the rate dropped to 8 mol CH₄ m⁻³ d⁻¹ during the dry season. The activity of methanotrophs is inhibited when the moisture content of the granular medium decreases below its field capacity, hence the frequency of irrigation should be increased as roots grow deeper and if plant species have uniform or parabolic root architecture (Abichou et al., 2011; Feng et al., 2017). Flux rate may also determine whether there are plant effects. Ndanga et al. (2015) conducted both laboratory and field experiments simulating the CH₄ oxidation behaviour of biosystems and found no difference in the CH₄ oxidation performance of vegetated and

non-vegetated biosystems up to a flux rate of about 100 g CH₄ m⁻² d⁻¹. However, they observed slight increases in CH₄ oxidation performance in vegetative systems at higher flux rates and even higher oxidation efficiencies in the bare-soil system. The highest CH₄ flux rates used were 270 and 180 g CH₄ m⁻² d⁻¹ for laboratory and field experiments respectively, representing the rates potentially observed at LBCs. Typically, bio-windows and methane biofilters are subjected to higher flux rates.

Despite the attempts of several researchers to incorporate vegetation in their soil methanotrophy research (Bohn et al., 2011; Reichenauer et al., 2011; Stralis-Pavese et al., 2004; Tanthachoon et al., 2008; Wang et al., 2008), most have not investigated the effects of root morphology (i.e., root length, dry biomass, and shape) on CH₄ oxidation. However, there has been considerable research on the contributions, plants make in transmitting gaseous compounds from various types of soils. Aerenchyma tissue in plants may provide a low resistance internal pathway for gas to transport between roots and shoot extremities (Bhullar et al., 2013; Colmer, 2003; Turner et al., 2020). Aerenchyma tissue promotes venting of gases such as ethylene, CO₂ and CH₄ produced by microbes in wet or flooded soil (Smith and Restall, 1971; Smith and Russell, 1969). The emission of CH₄ from wetlands can occur in the form of gas-bubble ebullition, molecular diffusion, and/or through plant-mediated transportation, where the gas diffuses through the aerenchyma tissues in vascular plants (Turner et al., 2020). Plant mediated CH₄ emissions, transported between the rhizosphere and the atmosphere account for about 64%–90% of the net CH₄ emissions from peatland soils (Shannon et al., 1996). A parametric study conducted by Feng et al. (2017) to determine the effects of root architecture, such as uniform, parabolic, exponential, and triangular shapes, on CH₄ oxidation found that plants with exponential, triangular root shapes and shorter root lengths increased CH₄ oxidation efficiency.

Although some work has been done, overall, there is insufficient understanding of the effects of plant root morphology on changes in gas profiles, moisture profile, methanotrophic bacterial populations and thereby biological CH₄ oxidation in granular media of field established LBCs, bio-windows and methane biofilters. Past research has failed to provide a conclusive answer to the question; what is the effect of vegetation on CH₄ oxidation? Therefore, the primary objective of the current study is to determine positive or negative effects of vegetation on CH₄ oxidation in LBCs, bio-windows and methane biofilters constructed in cold climates such as Alberta, Canada. The study answers several inter-related research questions: what are the impacts of vegetation on CH₄ oxidation under typical flux rates encountered at LBCs and at higher flux rates encountered at bio-windows and methane biofilters? What are the effects of different plant root morphologies on gas profiles and methanotrophic bacterial types and populations?

2. Materials and methods

Since lab-based study results cannot be extrapolated to CH₄ oxidation behavior in cold climates, this study was conducted outdoors simulating field conditions. The CH₄ oxidation potentials, gas profiles, methanotrophic populations and plant growth of three types of vegetated flow-through columns were compared with those of bare-soil columns.

2.1. Flow through column experiments

In order to achieve the study objectives, passively aerated flow-through column experiments were conducted. Passive flow-through columns established outdoors with plants represent the one-dimensional gas flow and CH₄ oxidation behavior of LBCs, bio-windows and methane biofilters.

2.1.1. Column set-up

Eight columns constructed from Plexiglas tubes with a 15 cm inner

diameter and 6 mm thickness were used for the flow-through column experiments. The schematic diagram of the columns and the eight flow-through experimental columns installed outdoors are shown in Fig. 1b. One end of each column was closed with an acrylic end-cap fitted with a rubber O-ring while the other end was left open to the atmosphere. Since the majority of microbial CH_4 oxidation is expected to occur within the top aerated zone of the system, the flow-through columns used in this work were designed with a total height of 57.5 cm. A mesh plate was placed between the gravel layer and packing material to prevent the cover soil/compost from clogging the distribution layer. Gas sample ports were drilled at 5 cm intervals down the core with stainless steel NPT fittings, male connectors and silicone septa. Fig. 1a is a schematic diagram of the flow-through columns used in this study.

Each column was layered with 7.5 cm of 16/32 gravel for gas distribution, 18 cm of topsoil (consisting of 40% sand, 34.2% silt, 16.1% clay and 9.7% gravel), and 32 cm of locally produced (yard and garden waste) compost mixture (compost: topsoil, 3:7 v/v) as packing granular material. The physical and chemical characteristics of the granular material are shown in Table 1. Moisture content (ASTM D 2216-98, 1998), bulk density (ASTM D7263-09, 2009) and pH (ASTM D4972-18, 2018) were determined for all of the granular materials.

2.1.2. Experimental procedure and schedule

The eight flow-through columns were set up outdoors at the Eco-center located in Okotoks, Alberta, Canada (50°42'58.78"N and 113°57'0.94"W). Duplicate columns were set up with three pairs containing vegetative columns and one pair containing only granular media with no vegetation. The vegetated columns consisted of Alfalfa (*Medicago sativa*), Canola (*Brassica napus*) and native grass blend (25% Northern wheat grass - *Elymus lanceolatus*, 35% awned wheat grass - *Elymus trachycaulus*, 10% June grass - *Koeleria macrantha*, 25% tufted hair grass - *Deschampsia caespitosa* and 5% rough fescue - *Festuca hallii*). All eight columns were irrigated at a rate of 400 mL week⁻¹. The selected vegetation types represent three different root morphologies. Furthermore, there is recent interest in growing cash crops, such as canola and alfalfa, on completed landfills to utilize the space in a productive manner (Battiston, 2013).

The experiments were conducted during the spring/summer-growing season of Alberta; from May 20 to September 4. The chronology of events is summarized in Table 2. The seed germination and plant establishment phase took place from May 20 to June 23 when the average daytime high ambient temperature was +21.2 °C, and nighttime lowest temperature was +1.5 °C. The flow-through column

Table 1

Physical characteristics of media used in the columns.

Media Characteristics	Media Type	
	Compost Mix (30:70)	Topsoil
Dry bulk density (g/cm ³)	0.989 ± 0.019	1.250 ± 0.025
Field capacity (w/w%)	88.97	44.16
pH	6.89 ± 0.01	7.13 ± 0.01
Moisture content (w/w%)	0.35	0.28

Table 2

Chronology of events.

Dates	Day	Activity
May 20 – June 23		Seed germination/plant establishment phase
June 24	0	Beginning of methane gas treatment
June 24 – Aug 21	0–11	Stage I gas treatment (180 CH ₄ /m ² /d)
	11–24	Stage II gas treatment (385 CH ₄ /m ² /d)
	24–50	Stage III gas treatment (575 CH ₄ /m ² /d)
	50–58	Stage IV gas treatment (815 CH ₄ /m ² /d)
	58–72	Stage V gas treatment (960 CH ₄ /m ² /d)
Aug 21 – Sep 4	72	Dismantling of columns and sampling

experiments were conducted from June 24 to August 20, when CH_4 gas (average 92% pure) was fed through the bottom of the columns at increasing flow rates in 4 stages. During this 58-day time period, the lowest ambient temperature was +5 °C and the highest was +33 °C. In Stage I, the gas flux rate was maintained at 180 g CH_4 m⁻² d⁻¹, representative of the flux rates observed at landfills accepting biodegradable organic waste. From stage II to IV, the flux rate was increased in a stepwise manner. Typically, higher flux rates, as tested here, are observed at bio-windows and methane biofilters established at landfills with semi and full landfill gas collection systems, respectively.

During Stage V, the CH_4 influx was increased to 960 g CH_4 m⁻² d⁻¹ only for the vegetated columns for another 15 days (day 58–72) to differentiate between the oxidation performances of the three different vegetated columns. It was necessary to conduct this phase of the experiment because there was no statistically significant difference between the performances of the vegetated columns at day 58.

2.1.3. Data collection and analysis

The eight flow-through columns were monitored over a period of 72 days to determine the comparative performance, and to screen key diagnostic parameters that could lead to improving the predictability of

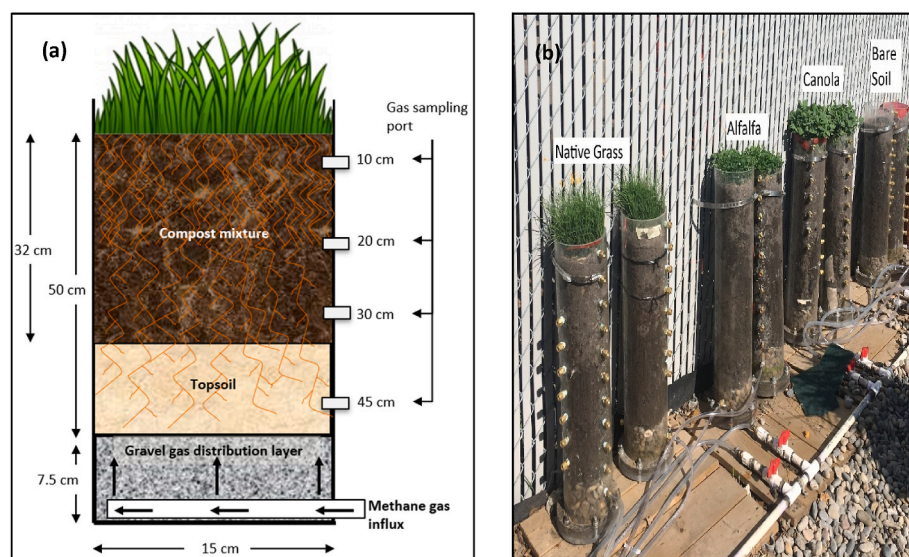


Fig. 1. a) A Schematic diagram of the flow-through columns; b) The outdoor set-up of experimental columns.

CH₄ oxidation performance. Oxidation efficiencies were measured weekly, as were gas concentrations along the columns. Sampling of the media occurred after 72 days in one of the duplicate columns for the four column types, to measure the kinetic parameters and to perform the microbial analysis.

The static chamber technique was used to measure CH₄ surface flux by connecting a cylindrical flux chamber (diameter: 15 cm, height: 25 cm) with a Landtec GEM™ 5000 portable gas analyser (Irandoost, 2020). The total flux of CH₄ was calculated by multiplying the change of gas concentration over time with the chamber volume/cross sectional area ratio. Methane Oxidation Efficiency (MOE) and oxidation rate were calculated and used to compare the performances of the vegetated and bare-soil columns. MOE was calculated using the mass balance method as follows (Powelson et al., 2006):

$$MOE = \frac{(Flux_{loaded} - Flux_{emitted})}{Flux_{loaded}} (100) \quad (1)$$

where MOE is expressed as CH₄ oxidation efficiency, and Flux_{loaded} (in g CH₄ m⁻² d⁻¹) and Flux_{emitted} (in g CH₄ m⁻² d⁻¹) were the CH₄ inlet and surface emission fluxes of the columns, respectively.

To develop the depth-wise gas concentration profiles along the columns, 2 mL gas samples were taken from each sample port located at 10, 20, 30 and 45 cm depths and analyzed for CH₄, CO₂, O₂, and N₂ using an SRI 8610C (SRI Instruments, California, USA) Gas Chromatograph with a thermal conductivity detector. The SRI-GC was outfitted with two columns including a 1.8 m MolSieve and a 1.8 m HayeSep D. The initial oven temp of 50 °C was held for 4 min and then ramped at 20 °C per min to 140 °C. The initial pressure was set at 3 psi and then increased to 17 psi for 2.5 min followed by a final increase to 31 psi and held for 5.98 min. The various gas concentrations were determined by comparison to standard gases obtained from Praxair gases (Praxair, Calgary, Canada).

2.2. Incubation experiments to determine reaction kinetic parameters

After 72 days of continuous operation, the columns were dismantled; homogenized media samples of approximately 10 g from 0 to 15, 15–30 & 30–45 cm depths of each column were collected and placed in 250 mL sealed amber serum bottles. They were injected with CH₄ gas to achieve an initial headspace CH₄ ratio of 10% (v/v). The headspace CH₄ concentrations were measured in timed intervals (every 24 h) until the concentration reached zero. The gas samples were analyzed using gas chromatography for CH₄, CO₂, and O₂. To determine the maximum rate of CH₄ oxidation or V_{max}, CH₄ oxidation rates were plotted in substrate saturation curves as a function of the initial CH₄ concentration in the headspace and then the data was linearized with Lineweaver Burk plots (Pokhrel, 2006).

2.3. Microbiological analyses

Quantitative and qualitative microbiological assessments were conducted to determine whether the presence of plants influenced the quantities and the types of methanotrophs populating the granular media. Several composite samples from different depths (i.e., 0–15 and 15–30 cm) of the dismantled columns were collected to quantify the methanotrophic populations. DNA was extracted from 490 mg of column media using a Fast DNA SPIN Kit for soil (MP Biomedicals, Canada). In addition, a purification step was performed with 5.5 M guanidine thiocyanate (GTC) to ensure the purity and optimization of DNA yield (Knief et al., 2003). A q-PCR assay described by Kolb et al. (2003) was used to assess methanotroph populations based on the number of methanotroph-specific *pmoA* genes as shown in Table S1. Reactions contained 1 µM of each primer (A189-mb661), 1 µL of sample gDNA, 5.0 µL of SsoAdvanced Universal SYBR Green Supermix (Bio-Rad Laboratories, Montreal, Canada) and DNase-free water to make up to 10 µL (QIAGEN, Velno, Netherlands). Reactions were run on a Rotor-Gene

6000 (QIAGEN, Velno, Netherlands). Standards were created from a *pmoA* gene PCR-amplified from *Methylococcus capsulatus* Bath and cloned into a PJET 1.2 plasmid. The cured plasmid was quantified with a Qubit HS kit (Thermo-Fisher, Waltham, MA USA) and serially diluted from 10⁹ to 10² copies per µL to create a standard curve.

Illumina-based sequencing of bacterial 16S rRNA gene amplicons was performed as described by Sheremet et al. (2020). Briefly, the V3–V4 region of the 16S rRNA gene was amplified according to the Illumina protocol (Illumina, 2013) with the universal primers 341fw and 785r. Amplicons were pooled and sequenced on an Illumina MiSeq system (Illumina, Inc), and analyzed using QIIME2 version 2019.10 (Caporaso et al., 2010). Taxonomic assignment of amplicon sequence variants (ASVs) was performed with the feature-classifier plugin (Bokulich et al., 2018) employing a naïve Bayes classifier approach. The taxonomy classifier was trained on the SILVA database, release 132 (Quast et al., 2012).

2.4. Plant data

Since plant root biomass may play an important role in CH₄ oxidation efficiency of vegetated systems by creating sub-surface macro and micro-pores, for each plant type, the root biomass density and root length were measured at the end of the experiment (day 72). To determine dry-weight (the dry biomass density is expressed in terms of unit soil area), plants below the ground in an area of 4 cm² were clipped at 1 cm above ground and the root section was washed, placed in aluminum weighing dishes, weighed and oven dried at 80 °C for 24–48 h cooled in a desiccator and reweighed until the moisture was eliminated. The belowground biomass density was calculated based on equation (2) shown below.

$$Drybiomassdensity = \frac{Dryweight\ of\ belowground\ plant\ roots}{plotarea} \quad (2)$$

To measure the average root length of plants, several plants were removed from the soil/media matrix with extra caution to minimize root damage. Then, the length from soil level to the tip of the root was measured using a measuring tape.

2.5. Statistical analysis

A two-way analysis of variance (ANOVA) with post-hoc Tukey test was carried out to determine the significance of the different types of columns (vegetated and bare soil) and CH₄ influx at different stages on CH₄ oxidation rates in the column experiment. A two-way ANOVA was performed to validate the methanotrophic bacterial count of each column at a given depth (i.e. 0–15 cm and 15–30 cm). All calculations were performed using SigmaPlot 14.0 (Systat Software Inc.).

3. Results and discussion

3.1. Methane oxidation rates and efficiencies

The average CH₄ oxidation rates and oxidation efficiencies of vegetated and bare-soil flow-through columns during each of the first four stages of the experiment are presented in Table S2 provided in the Supporting Information section. The time-dependent oxidation rates determined as averages of the duplicate columns at each monitoring event of the columns are presented in Fig. 2.

As evident from Fig. 2, there was a marked difference in the CH₄ oxidation rates of vegetated and bare-soil columns. The bare-soil columns showed the highest average CH₄ oxidation rate of 455 g CH₄ m⁻² d⁻¹ on day 58 when the inlet flux was 815 g CH₄ m⁻² d⁻¹. The corresponding maximum CH₄ oxidation rates (in g CH₄ m⁻² d⁻¹) for native grass, alfalfa and canola columns were 170, 147, and 161, respectively.

According to Table S2, when the inlet flux was 180 g CH₄ m⁻² d⁻¹, from day 0–11, the average CH₄ oxidation efficiency of the bare-soil

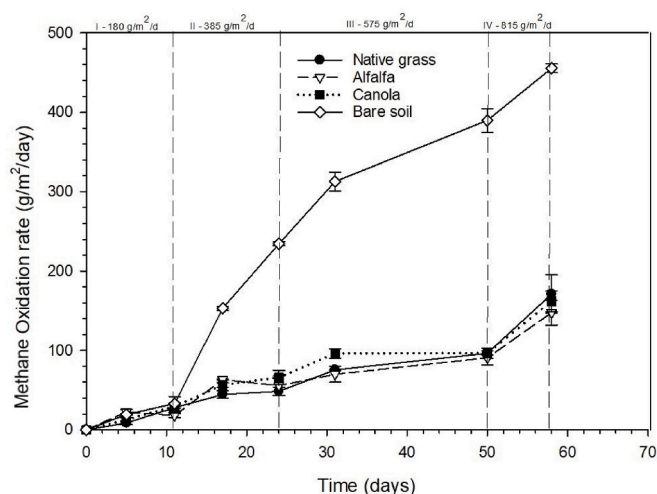


Fig. 2. Time-dependent methane oxidation rates (average of the duplicates) of vegetated and bare-soil flow-through columns.

columns was 15%. The corresponding maximum oxidation efficiencies of native grass, alfalfa, and canola columns were slightly lower at 12%, 12%, and 13%, respectively. However, the two-way Analysis of Variance (ANOVA) test showed that the differences in CH_4 oxidation among the different column types were not statistically significant ($p > 0.05$) at a loading rate of $180 \text{ g CH}_4 \text{ m}^{-2} \text{ d}^{-1}$. The differences of CH_4 oxidation efficiencies between the vegetated and bare-soil columns became apparent as the CH_4 influx was increased to $385 \text{ g CH}_4 \text{ m}^{-2} \text{ d}^{-1}$ during stage II, where the average oxidation efficiency in the bare-soil column was 50% and oxidation efficiencies were much lower in the vegetated columns with values 12%, 15% and 16% respectively for native grass, alfalfa, and canola. ANOVA showed that at the loading rate of $385 \text{ g CH}_4 \text{ m}^{-2} \text{ d}^{-1}$ the statistical significance (p -value) for the performance of bare-soil vs native grass was $p = 0.007$, bare-soil vs alfalfa was $p = 0.011$, and bare soil vs canola was $p = 0.01$. This indicates a significant difference in oxidation rates of the bare-soil column when compared to the three vegetated columns at loading rates beyond $385 \text{ g CH}_4 \text{ m}^{-2} \text{ d}^{-1}$. ANOVA also showed that there was no difference between the performances of the three vegetated columns at the loading rate $385 \text{ g CH}_4 \text{ m}^{-2} \text{ d}^{-1}$. Similar observations were made at higher CH_4 loading rates, with $p < 0.001$ for bare-soil vs native grass, $p < 0.001$ for bare-soil vs alfalfa and $p < 0.001$ for bare-soil vs canola at $575 \text{ g CH}_4 \text{ m}^{-2} \text{ d}^{-1}$ and $p < 0.001$ for bare-soil vs native grass, $p < 0.001$ for bare-soil vs alfalfa, and $p < 0.001$ for bare-soil vs canola at $815 \text{ g CH}_4 \text{ m}^{-2} \text{ d}^{-1}$. Differences between the 3 vegetated columns were not observed at CH_4 loading stages I to IV ($p > 0.05$), indicating no significant differences in oxidation rates among the 3 different vegetation column types.

Previous vegetated column/lysimeter experiments were conducted at relatively low input flux rates of ($100\text{--}150 \text{ g CH}_4 \text{ m}^{-2} \text{ d}^{-1}$ and lesser) and therefore it is not surprising that they reported no difference between the CH_4 oxidation activities of the vegetated and non-vegetated systems (Wang et al., 2008). Ndanga et al. (2015) also reported no difference in performance between vegetated and bare-soil laboratory columns and field established lysimeters up to an input flux rate of $100 \text{ g CH}_4 \text{ m}^{-2} \text{ d}^{-1}$. However, they observed statistically significant differences once the influx rates were increased beyond $100 \text{ g CH}_4 \text{ m}^{-2} \text{ d}^{-1}$, although the average deviation between oxidation efficiencies of the vegetated and bare-soil columns were only about 4%. It should be noted that the highest flux rate they used was $270 \text{ g CH}_4 \text{ m}^{-2} \text{ d}^{-1}$, much lower than the rates used in the current study.

The average oxidation rates of the vegetated columns at a loading rate of $960 \text{ g CH}_4 \text{ m}^{-2} \text{ d}^{-1}$ (day 58–72) are shown in Fig. 3. At this loading rate, the average oxidation rates of native grass, alfalfa, and canola columns were $235 \text{ g CH}_4 \text{ m}^{-2} \text{ d}^{-1}$, $125 \text{ g CH}_4 \text{ m}^{-2} \text{ d}^{-1}$, and $196 \text{ g CH}_4 \text{ m}^{-2} \text{ d}^{-1}$, respectively.

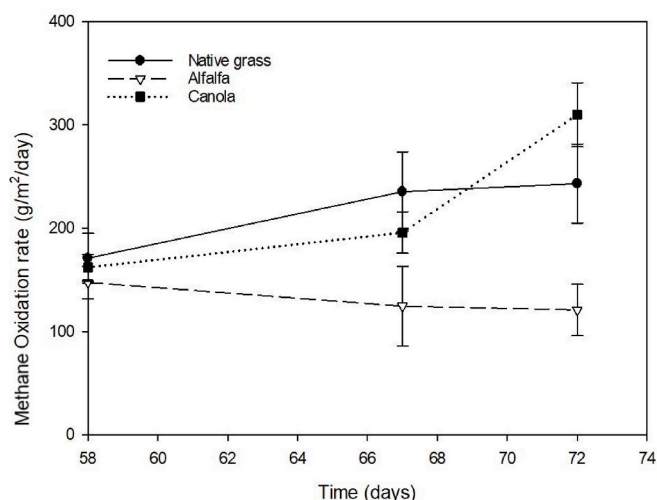


Fig. 3. Methane oxidation rates of the vegetated flow-through columns at $960 \text{ g CH}_4 \text{ m}^{-2} \text{ d}^{-1}$ from day 58 to day 72.

$\text{CH}_4 \text{ m}^{-2} \text{ d}^{-1}$ respectively on day 67, and $243 \text{ g CH}_4 \text{ m}^{-2} \text{ d}^{-1}$, $121 \text{ g CH}_4 \text{ m}^{-2} \text{ d}^{-1}$, and $309 \text{ g CH}_4 \text{ m}^{-2} \text{ d}^{-1}$ respectively on day 72. The statistical significance as indicated by ANOVA for canola vs alfalfa was $p = 0.009$, native grass vs alfalfa was $p = 0.012$, and canola vs native grass was $p = 0.718$.

There are a number of potential reasons for this observation, and one of them could be the presence of nitrogen-fixing root nodules in leguminous plants such as alfalfa, where the plant hosts bacteria that converts atmospheric N_2 to NH_4^+ (Bodelier and Laanbroek, 2004). Reay et al. (2005) observed a reduction in CH_4 oxidation in soil containing alder plants (a symbiotic non-legume nitrogen fixer) due to the inhibitory effects of elevated NH_4^+ and NO_2^- which is produced during NH_4^+ oxidation.

3.2. Gas concentration profiles

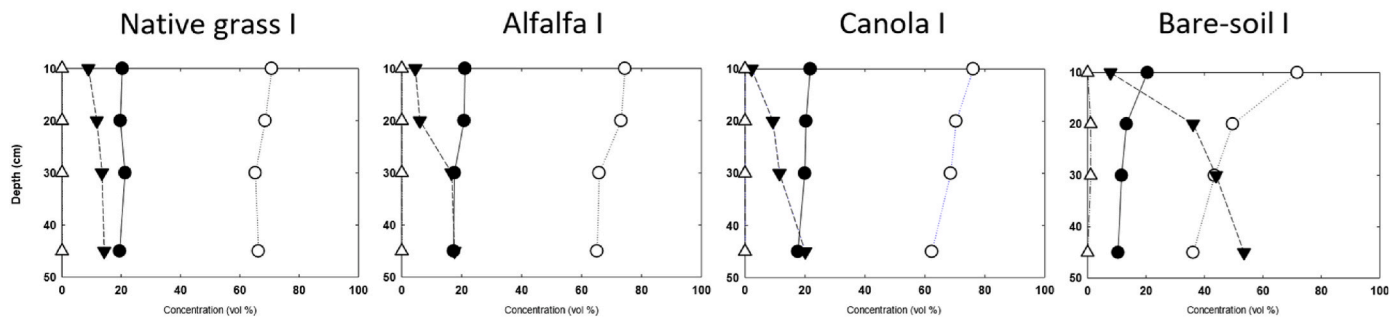
Fig. 4 includes representative gas profiles at different depths of four vegetated and bare-soil columns during the four stages of the study.

Fig. 4a shows that on day 8 (during stage I of the study), the columns with vegetation exhibited higher atmospheric air penetration relative to bare-soil columns. The N_2 concentrations ranged between 71% and 66% in native grass, 74% and 65% in alfalfa, and 76% and 62% in canola at depths of 10 cm and 45 cm, respectively. On the other hand, the N_2 concentrations in the bare-soil column were 72% and 36% at depths of 10 and 45 cm.

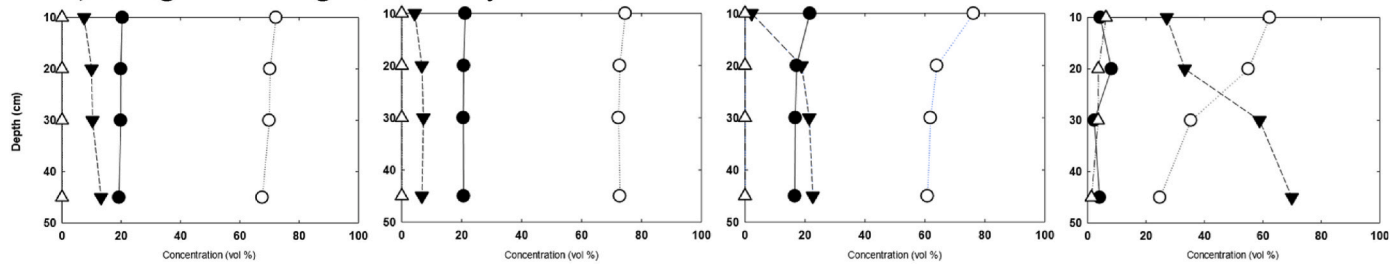
The O_2 concentrations also showed similar trends with relatively high levels at 45 cm depth in vegetated columns compared with the bare-soil column. Concentrations of CO_2 were only elevated in the bare-soil column with a concentration of 1% at depths of 20 and 30 cm consistent with high levels of CH_4 oxidation. Furthermore, the columns with vegetation showed significantly lower CH_4 concentrations compared to the bare-soil column, with concentrations ranging from 9% to 14% in native grass, 4%–18% in alfalfa, 2%–20% in canola, and 8%–53% in bare soil columns at depths of 10 cm and 45 cm.

On day 24 (during stage II), the differences in atmospheric air penetration and aeration were more pronounced than on day 8 (see Fig. 4b). The O_2 concentrations ranged between 20% and 19% in native grass, 21% throughout the column in alfalfa and 22%–17% in canola, while the bare-soil column had significantly lower concentrations of 4%, 8%, 2%, and 4% at depths of 10 cm, 20 cm, 30 cm, and 45 cm, respectively. Despite the increase in the CH_4 loading rate and corresponding high rates of CH_4 oxidation in stage II, CO_2 was only detected in the bare-soil column with 6%, 4%, 4%, and 1% at 10 cm, 20 cm, 30 cm, and 45 cm depths, respectively.

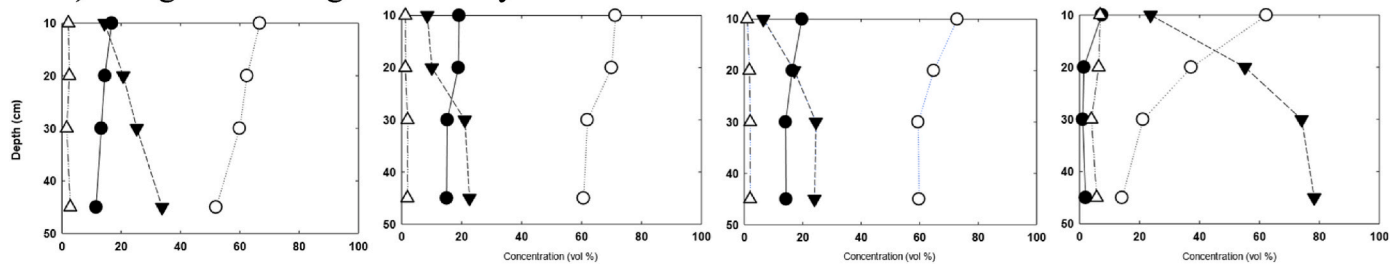
a) Stage I – 180g/m²/d – Day 8 Profile



b) Stage II – 385g/m²/d – Day 24 Profile



c) Stage III – 575g/m²/d – Day 31 Profile



d) Stage IV – 815g/m²/d – Day 58 Profile

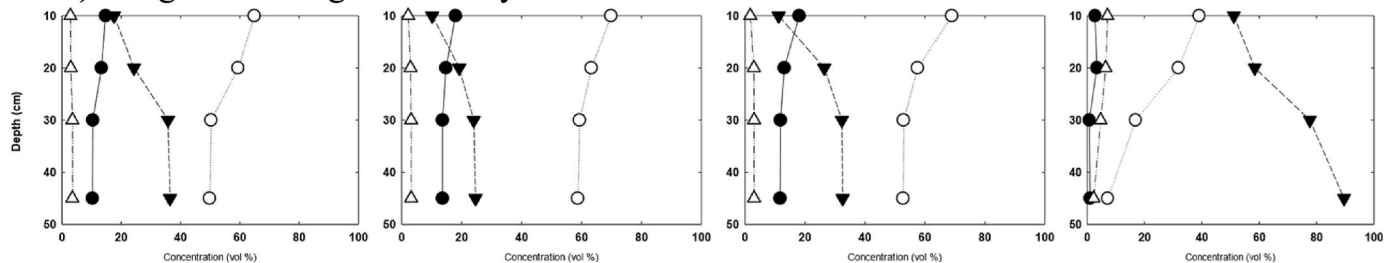


Fig. 4. Gas profiles at different depths at different stages of Native grass I, Alfalfa I, Canola I and Bare soil I columns (Symbols; —●—: O₂, —○—: N₂, —▼—: CH₄, and —△—: CO₂).

In stages III and IV (see Fig. 4c and d), the gas concentration profile trends continued with major differences between the vegetated and bare-soil columns. Low concentrations of CO₂ were detected for the first time in the vegetated columns during stage III (Fig. 4c) with concentrations ranging around 1–2% along the different depths, while the values were higher in the bare-soil column at 4–7%. The CO₂ concentrations detected in stage IV were significantly higher in all columns when compared to previous stages, with the highest values being

exhibited in the bare-soil column, corresponding with high CH₄ oxidation rates. The CO₂ concentrations ranged from 2 to 3% in columns with plants and 2–7% in bare soil columns (Fig. 4d). During stage IV, some differences could be observed between the O₂ concentration profiles of vegetated columns. The alfalfa vegetated column showed the highest O₂ concentration at a depth of 45 cm. This observation is consistent with the root penetration of vegetated columns with plants with long roots encouraging atmospheric air penetration and subsequent low CH₄

oxidation.

Previous studies also reported significantly higher O_2 concentrations throughout vegetated columns and lysimeters compared with bare-soil systems (Reichenauer et al., 2011). Researchers have postulated that plant root systems improve soil diffusivity and physical properties controlling diffusion of atmospheric air deep into soil (Bohn et al., 2011; Stralis-Pavese et al., 2004; Reichenauer et al., 2011; Chai et al., 2010). Soil diffusivity experiments conducted by Bohn et al. (2011) showed a high diffusion coefficient in the root zone area of the column vegetated with goldenrod. The increased diffusion coefficient improves the O_2 supply required by methanotrophs to oxidize CH_4 in deeper layers.

In addition to diffusion coefficient of the media, other factors contribute to aeration of columns and CH_4 oxidation potential. Plants with aerenchymatous structures are known to act as gas conduits and play a major role in diel methane emissions and can account for up to 90% of the total CH_4 efflux into the atmosphere (Xin et al., 2016; Ding et al., 2004). Researchers have speculated that plant root systems could potentially create preferential pathways that can negatively impact CH_4 oxidation (Ndanga et al., 2015). These preferential pathways not only create routes for the CH_4 to escape but also increased porosity allowing free movement of gas, reducing the retention time of CH_4 within the column, thereby hindering the CH_4 oxidation process performed by methanotrophs. This was the case in our experiments where high concentrations of both CH_4 and CO_2 were observed throughout bare-soil columns while lower concentrations of CH_4 and CO_2 were observed in vegetated columns.

The plant data collected at the end of the experimental run provided information on the role root systems play in soil aeration and CH_4 oxidation. The average root lengths for the different plants in columns are shown in Fig. 5a. Alfalfa had the longest roots at 38 cm, followed by canola at 31.5 cm and native grass at 26.5 cm. As depicted in Fig. 5b, below ground root dry biomass density of alfalfa (0.46 g cm^{-2}) was almost double that of native grass (0.25 g cm^{-2}). Canola columns had a dry biomass density of 0.30 g cm^{-2} . The alfalfa columns with high dry biomass density increased soil diffusivity and thereby improving atmospheric air penetration deeper inside the column resulting in the highest O_2 and N_2 concentrations.

Our plant data provided further information on the physical and biological processes occurring in the columns. With long roots and high below ground biomass, alfalfa vegetated columns exhibited the highest O_2 and N_2 concentrations and the lowest CH_4 concentrations at the deepest depths. Consequently, alfalfa showed the lowest CH_4 oxidation rates due to the short-circuiting of CH_4 via the roots. High below ground biomass density also contributed to high aeration in the root zone but

also created preferential pathways for CH_4 gas to escape, further decreasing the CH_4 oxidation rate. Although increased aeration increases CH_4 oxidation potential, the benefit of aeration is counteracted by the creation of macropores and conduits for CH_4 gas short-circuiting. However, it is worth noting that these effects are mostly notable at high CH_4 loading rates.

Research conducted by others to study plant-mediated CH_4 emissions has shown that vascular plants influence the production, transportation (known as the conduit or chimney effect), and oxidation of CH_4 (Bhullar et al., 2013). An experiment conducted with forbs (an agricultural weed) and graminoids (grass like herbaceous plants) observed a positive correlation between plant mediated CH_4 emissions and the plant root volumes (Bhullar et al., 2013). Several studies done on rice paddy fields have also shown a positive correlation between CH_4 emissions and plant biomass (Chanton et al., 1997; Sass et al., 1990). Our study also provided clear evidence of short-circuiting of CH_4 due to plant roots that resulted in decreased CH_4 oxidation in alfalfa vegetated columns.

3.3. Kinetic parameters of methane oxidation

Typically, the average maximum rate of CH_4 oxidation (i.e., V_{max}) data provides a quantitative indication of the amount of methanotrophic activity within each column and the CH_4 oxidation potential in each section of the column. The average V_{max} values for various depths of the different columns are presented in Table 3.

The V_{max} values decreased consistently from top to bottom of each column, except in the column with native grass at a depth of 15–30 cm. However, there was somewhat better distribution of methanotrophic activity in the deeper layers of the vegetated columns than in bare-soil columns, with V_{max} in the deep layer (30–45 cm) of the canola column being about 13% of top layer. In bare-soil columns, the V_{max} of the deep layer was only 1% of the top layer. Somewhat similar results were observed by Lee et al. (2014) during a laboratory column study of LBCs where average V_{max} values of 8.40 , 7.11 , and $5.97 \mu\text{mol g}^{-1} \text{dw}^{-1} \text{h}^{-1}$

Table 3

V_{max} values (average) of media samples obtained at the different depths in vegetated and bare-soil columns.

Sample Depth (cm)	Kinetic parameter of the columns - V_{max} ($\mu\text{mol/g dw/h}$)			
	Native grass	Alfalfa	Canola	Bare soil
0–15	3.36	2.65	6.87	15.38
15–30	1.74	0.86	1.02	10.96
30–45	1.82	0.28	0.92	0.15

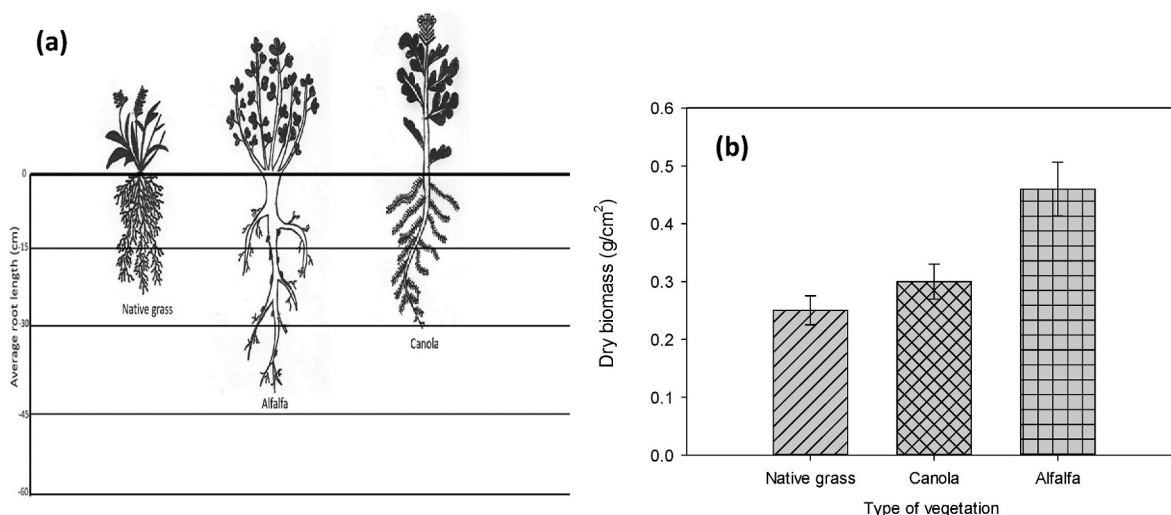


Fig. 5. a) Average root lengths of the vegetated columns on day 72; b) Below ground dry biomass density of the vegetated columns on day 72.

were observed in the top (0–10 cm), middle (10–40 cm), and bottom (40–50 cm) layers, respectively. The general consensus among researchers is that most CH_4 oxidation occurs in the top layer of an LBC due to the inability of surface O_2 to diffuse deep into the soil (Gebert et al., 2011). However, our results demonstrated that vegetated columns exhibit significantly lower activity than bare-soil columns, with the lowest being displayed in the alfalfa columns. This observation is compatible with our gas profile data in which differences in CH_4 concentrations in the gas profiles in vegetated and bare-soil columns were prominent. The ingress of atmospheric O_2 deep inside the columns did not translate into high overall methanotrophic activity, although there was activity within deep layers, albeit to a lesser extent than top layers. In comparison to other studies reported in the literature, the V_{max} values reported here are relatively high (particularly in bare-soil column). This is the result of using high CH_4 loading rates during latter stages of the study. A significant influence of CH_4 flux rates on methanotrophic activity was reported by Pawłowska and Stępniewski (2006). They observed a 1.1–2.5-fold increase in methanotrophic activity when the CH_4 concentration increased by 8-fold in the column.

3.4. Methanotrophic bacteria population

The characteristics of the methanotrophic populations within granular media exposed to high CH_4 concentrations provide an indication of the changes that may happen over time due to external influences, including changes in temperature, moisture, root biomass and associated changes to nutrient content and aeration. The quantification of methanotrophs and methanotrophic community analyses were performed after 72 days of continuous operation of the columns. The methanotrophic bacterial populations under vegetated and bare-soil columns at different depths are shown in Fig. 6a. The quantities of methanotrophs are expressed in terms of number of *pmoA* gene copies found in 1 g of the media tested. In the shallow depths (0–15 cm) of native grass, alfalfa, canola, and bare-soil columns, 2×10^4 , 1.5×10^3 , 4.4×10^5 , and 1.1×10^7 *pmoA* gene copies were found, respectively, while in the deeper depths (15–30 cm) the numbers were 3.8×10^5 , 2.8×10^4 , 1.4×10^6 , and 1.1×10^7 , respectively. The presence of vegetation significantly decreased the methanotrophic bacterial populations in the flow-through columns, both in the shallow and deep layers. As mentioned earlier, the concentration of CH_4 significantly influences methanotrophic activity and the methanotrophic bacterial count. The significantly higher CH_4 concentrations throughout bare-soil columns

when compared with vegetated columns led to the orders of magnitude differences in the methanotrophic bacterial count. The methanotrophic population in the bare-soil columns were significantly higher ($p < 0.001$) at the different depths (0–15 cm and 15–30 cm) when compared to all vegetated columns.

The 16S rRNA gene amplicon data was searched for known methanotrophic taxa within the *Gammaproteobacteria* (family *Methylococcaceae*) and *Alphaproteobacteria* (families *Methylocystaceae* and *Beijerinckiaceae*). Known methanotrophic taxa were present in all samples. The relative abundance of *Gammaproteobacteria* methanotrophs (sometimes called Type I methanotrophs) versus *Alphaproteobacteria* methanotrophs (Type II) in vegetated and bare-soil columns at shallow (0–15 cm) and deep (15–30 cm) layers is shown in Fig. 6b. *Gammaproteobacteria* methanotrophs were the most abundant in both the shallow and deeper depths of all columns, making up >80% of all methanotrophic reads. Smaller proportions of *Alphaproteobacteria* methanotrophs were found in deeper (15–30 cm) layers of alfalfa (4% of total), canola (20%) and bare-soil (10.9%) columns. Typically, *Gammaproteobacteria* methanotrophs with relatively low CH_4 oxidation capacity tend to favour low CH_4 (<1000 ppmv) and high O_2 concentrations while Type II methanotrophs with high CH_4 oxidation capacity prefer high CH_4 (1000–10,000 ppmv) and low O_2 concentrations (Hettiarachchi et al., 2007; Karthikeyan et al., 2015). Our results do not provide conclusive evidence on the effect of vegetation and related changes to aeration on the types of methanotrophs in columns. Unfortunately, we were not able to determine time-dependent changes to bacterial populations since the bacterial populations were only determined at the end of the experiment after 72 days of continuous operation.

3.5. Moisture profiles

The moisture content of the granular medium supporting methanotroph growth is an important factor affecting CH_4 oxidation potential. The expectation is that vegetation will impact media moisture content. The gravimetric water contents measured at the end of the experiment at depths of 0–15, 15–30, and 30–45 cm in each column are shown in Table 4. Evidently, moisture content values along the depths in the bare soil column were much higher when compared with the vegetated columns possibly due to water loss from evapotranspiration in vegetated columns. Even among the vegetated columns, the moisture content profiles exhibited different patterns due to differing root lengths and biomass densities. With deep roots, alfalfa played a major role in

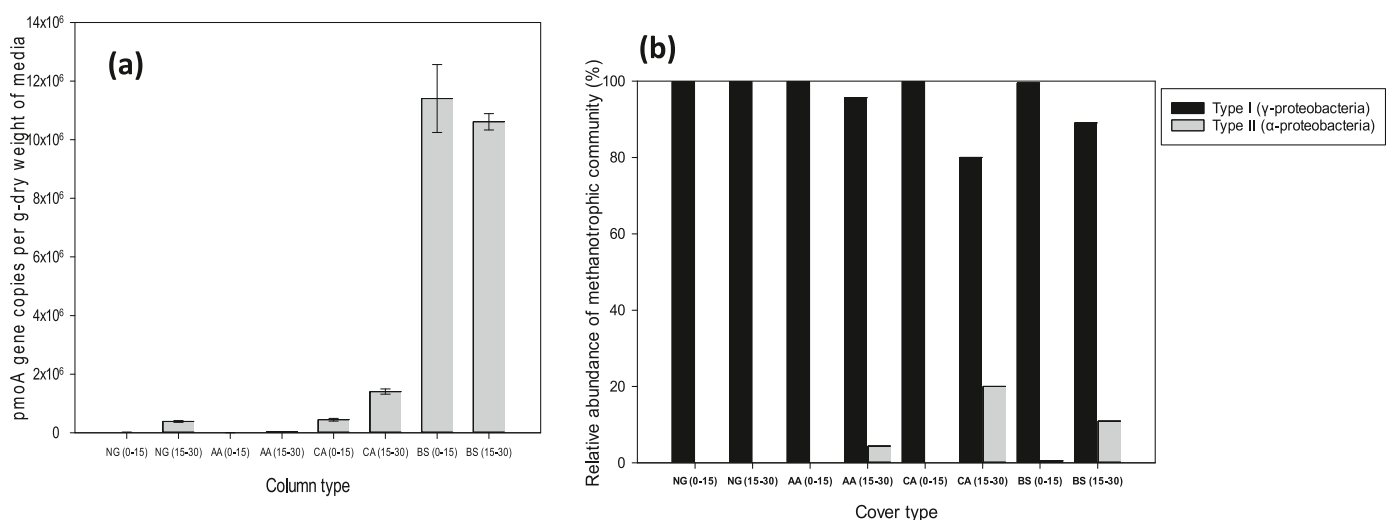


Fig. 6. a) Comparison of the *pmoA* gene copies in different columns in shallow and deep layers on day 72; b). Relative abundance of Type I and II methanotrophs on day 72 (γ -proteobacteria versus α -proteobacteria) in vegetated and bare soil columns. Note: NG: native grass, AA: alfalfa, CA: canola, BS: bare-soil, at the depths of 0–15 cm and 15–30 cm.

Table 4

Depth profiles of moisture content at the end of the experiment.

Type of vegetation	Depth (cm)	Water content % (w/w) dry basis
Native grass	0–15	28.20
Native grass	15–30	25.06
Native grass	30–45	29.12
Alfalfa	0–15	25.62
Alfalfa	15–30	21.59
Alfalfa	30–45	19.96
Canola	0–15	27.18
Canola	15–30	19.49
Canola	30–45	28.35
Bare soil	0–15	28.23
Bare soil	15–30	32.09
Bare soil	30–45	39.15

decreasing moisture content at the 30–45 cm layer. Since their roots did not reach deeper layers of the columns, the lowest moisture contents in canola and native grass columns were observed at the 15–30 cm layer. Jalilzadeh (2019) conducted a field lysimeter study in the same location and also observed changes in volumetric moisture content down to a depth of 20–30 cm in grass-covered lysimeters and down to a depth of 50–60 cm in alfalfa covered lysimeters. They correlated these observations to evapotranspiration losses. Reichenauer et al. (2011), also reported higher water content in unplanted lysimeters when compared with the planted lysimeters. They observed that lysimeter planted with both alfalfa and grass had slightly lower water content at deeper layers when compared to poplar, miscanthus and grass. Tanthachoon et al. (2008) reported that the reduction in moisture content due to vegetation, especially under non-irrigated conditions, can cause microbial water stress and lower microbial activity. Visvanathan et al. (1999) reported negligible CH₄ oxidation when moisture content was as low as 6%. In our study, none of the moisture content profiles dropped to the levels cited in literature that could have an inhibitory effect. Visvanathan et al. (1999) also reported optimum moisture content of 15%–20% for maximum CH₄ oxidation. All our columns exhibited moisture contents reasonably close to the optimum levels indicating that moisture content should not be considered an influencing factor of CH₄ oxidation performance in this study. In fact, water was added during the experiment to maintain the moisture content at consistent optimum level.

4. Conclusion

This study was conducted to investigate the effects of vegetation on microbial CH₄ oxidation potential of engineered biosystems, including landfill biocover systems (LBCs), bio-windows and methane biofilters, using outdoor flow-through columns. The gas and moisture profiles, CH₄ oxidation kinetic parameter values, methanotrophic bacterial count, and methanotrophic bacteria type along the column depths supported the CH₄ oxidation data gathered from experiments conducted on vegetated and bare soil flow-through columns. Overall, the results indicated a significantly negative impact of vegetation on microbial CH₄ oxidation.

The maximum CH₄ oxidation rate ($V_{\max}$) at 0–30 cm depths for bare-soil columns was several folds higher than the vegetated columns, similar trends were observed in methanotrophic population levels with *pmoA* gene copies in the scale of $\sim 10^7$ for bare-soil columns, while the populations ranged between ~ 1500 to 10^5 for vegetated columns. Type II methanotrophs that thrive in high CH₄ and low O₂ concentrations were observed in deep layers of the alfalfa and canola columns and in both shallow and deep layers of bare soil columns in small amounts. These results agree well with the gas profile data that indicated higher CH₄ concentration in bare-soil columns and deeper levels of vegetated columns. However, Type I methanotrophs dominated throughout the columns both in vegetated and bare-soil columns. In addition, due to the

water uptake by the root systems of the plants the moisture content in vegetated columns was much lower than the bare-soil columns. However, the lowered moisture contents were not low enough to inhibit the microbial activity.

The influence of vegetation on CH₄ oxidation was not evident at relatively low CH₄ influx rates but was clear when the CH₄ influx was increased to 385 g CH₄ m⁻² d⁻¹. In typical sanitary landfills accepting biodegradable organic waste, the average CH₄ flux is in the range of 150 g CH₄ m⁻² d⁻¹. However, most of these emissions occur at “hot spots” and the majority of the landfill surface emits less than 100 g CH₄ m⁻² d⁻¹. Therefore, it is reasonable to conclude that vegetation does not impact CH₄ oxidation in most LBCs as long as they are designed and constructed to eliminate “hot spots.” Unlike LBCs, very high influx rates could be encountered at landfill bio-windows and methane biofilters. Plant growth could impact the CH₄ oxidation performance of such engineered biosystems. Nevertheless, the selection of plants with shallow roots and low root biomass below surface could minimize the impact.

In conducting flow-through column experiments, we utilized relatively short columns and media bed thicknesses because the interest was on the active CH₄ oxidation region of the media bed. However, plants with long roots penetrated the entire media bed and therefore may have amplified the conduit or chimney effect of roots. It is possible that plants grown on thick media beds may not show the drastic effects we observed at high influx rates.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.soilbio.2022.108583>.

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